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# Clinical Practice Guidelines for Assessment and Management of Children and Adolescents Presenting with Psychiatric Emergencies

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## INTRODUCTION

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India ranks second in being the most populous country. Every fifth person is an adolescent between 10 and 19 years, and every third person is aged between 10 and 24 years. We have a record of the highest number of children and adolescents, more than 434 million. There is wide variation in reporting of psychiatry disorders in CAMH (Child and Adolescent Mental health), and the recent national mental health survey reported a prevalence of 7.3% of morbidity among adolescents. There is no national data on the prevalence of psychiatric emergencies. Overall, there is a trend for increased utilization of psychiatry emergency services by children and adolescents.<sup>[1]</sup>

An American study reports a 60% increase in mental health disorder is the reason to visit emergency services.[2] Literature reports that the most frequent emergencies are suicidal behavior, depression, aggressiveness, substance abuse, and violence-related situations. The emergency may be related to an underlying mental health condition, which has relapsed or may be the first episode of an illness. Repeat visits to the emergency services have been reported from 20 to 47%.[3,4]

A psychiatric emergency is an acute disturbance of either behavior, thought, or mood of a person and has the potential for a negative impact. This emergency if unattended can cause harm to the patient or other community members. The clinical presentation of psychiatric emergencies in CAMH is usually different from that seen in adults. Assessment warrants identifying symptoms, assessment of underlying disorder, the impact of the emergency on the child and family, the protective factors, and the resources for management. Besides a good history, a thorough examination to rule out medical comorbid or etiological disorders is important. It is imperative to quickly triage and pinpoint symptoms requiring immediate attention. These could be psychomotor agitation, aggressiveness, violence, delirium, and suicidal behavior. Investigations such as drug screening, blood count, electrolyte analysis, cardiac monitoring, and computed tomography may be required in some cases. Early and prompt identification and treatment would yield positive results. An important competency required is the ability to assess risk and manage it effectively.

With this large child and adolescent population, coupled with a paucity of child mental health professionals to serve the psychiatry disorders in CAMH, it is important to have standard guidelines for managing psychiatric emergencies in CAMH. A guideline on assessment and management of psychiatric emergencies in children and adolescents encompasses identifying at-risk patients, ensuring safety, interviewing the child and parent, developing a therapeutic alliance, examination, and management of both pharmacological and nonpharmacological approaches strategies.

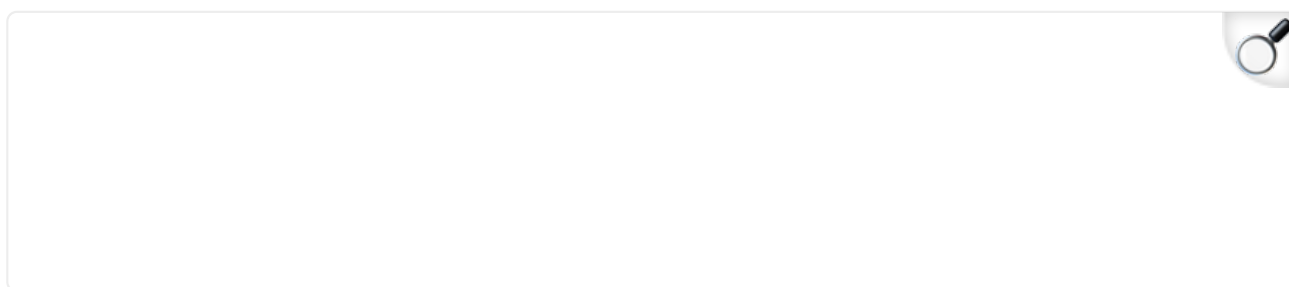
## EPIDEMOLOGY

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Children and adolescents visit the emergency for a variety of reasons. A western study reported that nearly three-fourths of the patients had a primary psychiatric diagnosis. The

most frequent diagnoses reported were anxiety states/panic disorder (14%), depression (13%), drug abuse (11%), and conduct disorders (8%). Other studies have reported personality disorders and schizophrenia/psychotic disorders, aggressive behavior, thoughts or actions of self-harm, medication refills, and autism spectrum disorders as reasons to visit the emergency services. Trends have shown a sharp rise in self-injury and suicide-related emergency visits among children and youth.[2] The common presentations can be seen in [Figure 1](#).

Figure 1.



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### Presentation to emergency services

Self-harm behaviors encompass suicide attempts, deliberate self-harm, and nonsuicidal self-injury (for key terms in suicide literature, refer to IPS CPG on assessment and management of suicidal behaviors). The World mental health report 2022 highlights suicide to be the third leading cause of death in 15–29 years and the fourth leading cause of death in males in this age group. Overall, it is the fourth leading cause of death among 15–29 years old and accounts for 8% of all deaths in this age group.

Aggression refers to a behavioral style aimed at deliberately harming other people or objects and is considered a way to adapt, but it can be abnormal when rules are broken. It is a common phenomenon and an important associated feature of psychiatric disorders affecting 10–20% of youth. “Terrible twos” refers to developmentally appropriate

aggression in toddlers, which peaks at 30 months and declines by 5 years of age due to the development of self-control and cognitive competencies. Social and relational aggressions are indirect forms of aggression seen in school-going children. During adolescence, the awareness of self-identity and social standing with peers, desire to fit in, and the desire for popularity can lead to greater aggression.

Acute confusional state, commonly known as “delirium” is characterized by an abrupt/acute onset of altered sensorium with a change/fluctuation in baseline mental status, inattention, disorganized thinking with or without perceptual abnormalities (delusions and/or hallucinations), and is the result of an underlying medical condition. It is a vastly underrecognized and underdiagnosed entity in children and adolescents and neither DSM 5 nor ICD 10/ICD 11 includes a definition of delirium specific to children and adolescents. Also, given the developmental stage of children, it becomes difficult to apply the definition of adult delirium as it is in children.<sup>[5]</sup> However, with the development and use of multiple validated tools to evaluate delirium in children and adolescents, it is being increasingly recognized and comprises 10% of all pediatric consultation–liaison referrals. It accounts for ~30% of referrals for critically ill children and is a marker for serious illness.<sup>[6]</sup> Like the types of adult delirium, delirium in children and adolescents is classified into three subtypes based on psychomotor state-hypoactive delirium (apathetic/uninterested child), hyperactive delirium (irritable, thrashing child), and mixed delirium (fluctuates between hypoactive and hyperactive state). While many clinical features of adult delirium may be applicable to children, yet some features are more prominent in children. These are irritability, agitation, affective lability, sleep-wake cycle disturbances, and fluctuations in symptoms. Perceptual abnormalities (delusions, hallucinations), speech disturbances, and memory impairments are less commonly seen in children. Acute onset developmental regression with loss of previously acquired skills, inconsolable child, and reduced eye contact with the caregiver are some unique features of delirium in young children.

Psychosis is defined as a disruption in the thought process, delusions (false, unshakeable beliefs), and hallucinations (false perceptions in the absence of an external stimulus) as a presenting complaint in children and adolescents in the emergency setup may be a manifestation of a primary psychiatric illness, substance withdrawal or intoxication, or may occur in the context of a medical condition.<sup>[7]</sup>

Among the various manifestations, anxiety symptoms/disorders are among the most common psychiatric conditions in children and adolescents and are associated with an increased risk of suicide attempts and significant morbidity and mortality. The course is considered chronic, persistent, and recurring with high levels of short-term and long-term impairment. At any given time, about 7% of youth worldwide have an anxiety disorder and are more common in girl. The lifetime prevalence rates among 13–18 years is approximately.

20% for specific phobia, 9% for social anxiety, 8% for separation anxiety, and 2% each for agoraphobia, panic, and generalized anxiety. Approximately 4% of children and adolescents experience posttraumatic stress disorder (PTSD) with increases seen in children exposed to trauma. Nearly one-quarter of adolescents presenting to the emergency services have been screened and found to have symptoms compatible with preexisting PTSD.[8] Occasionally, children present with an agitation which needs urgent intervention.

Substance use may present as intoxication, withdrawal, or dual diagnosis. Studies have found that substance abuse and mental health conditions presenting to the emergency have overlap with one in five visits for substances complicated by mental health comorbidity. Almost all mental health subcategory was positively associated with substance use.[9]

## ASSESSMENT OF CHILDREN AND ADOLESCENTS PRESENTING WITH PSYCHIATRIC EMERGENCIES

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### Development of a therapeutic alliance

This is the key to a comprehensive assessment. A solid alliance enables improved outcomes. A strong alliance includes understanding the patient and parents' priorities, cultural competence, and interpersonal warmth. When working with children or youth with mental illness or at risk of suicide, developing a strong working alliance with parents, caregivers is an important component of the overall relationship-building process. A therapeutic alliance is an interpersonal process that has relational, cognitive, and

emotional dimensions and goes beyond superficial friendliness. The task of the clinician is to reach, together with the person, a shared understanding of the person’s illness and suicidality.

The qualities associated with the development of a strong therapeutic alliance are – credibility, warmth, genuineness, empathy, flexibility, regular solicitation of feedback, and common understanding regarding treatment goals. Key aspects of a therapeutic relationship are highlighted in [Table 1](#).

Table 1.

Key aspects on therapeutic relationship

|                                                                                                                                  |
|----------------------------------------------------------------------------------------------------------------------------------|
| Key aspects of building therapeutic relationships                                                                                |
| Shared understanding of a young person’s thoughts, emotions, beliefs and suicidality/aggression                                  |
| Acknowledge emotional pain in all the patients and recognize that thoughts of suicide are understandable under the circumstances |
| Convey empathy and instill hope to young people and their parents/caregivers                                                     |
| Feedback                                                                                                                         |

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History taking and examination

The basic tenets of psychiatric assessment, i.e., thorough history taking, general physical examination along with a detailed neurological examination, mental status examination, and relevant investigations are to be followed while evaluating children and adolescents presenting with psychiatric emergency. The history should be collected from as many

sources as possible; this becomes important because of various reasons like the child may not have acquired speech and language skills, may be having neurodevelopmental conditions which prevent him from active communication, may not feel safe, secure, and confident to discuss stressful issues, or may be defiant and not ready to talk. It is essential to interview both the parent and child separately. Various techniques for gathering information can be employed, such as interview, play observation, and behavior observation. It is also essential to observe their interaction. This will help the clinician understand the genesis of the various issues and plan intervention. Parents and children may be able to share all concerns in the absence of the other. Children may be able to confide in the therapist in the absence of parents.[10]

History taking should include:

- All details of behavior under evaluation – onset, duration, frequency, precipitating/maintaining/relieving factors
- Birth and developmental history to assess for neurodevelopmental conditions like intellectual disability, attention deficit hyperactivity disorder, autism spectrum disorder, learning disability
- Substance use history
- Medication use – past and present
- Presence of constitutional symptoms like fever, headache, arthralgias, rash
- Academic performance
- Risk of self-harm
- Psychosocial information: family structure, family and peer relationships, living situation, any adverse situations/childhood experiences (like trauma and abuse), neighborhood environment, type of parenting
- Past history – hospitalizations, medical conditions like epilepsy, head injury, systemic illnesses
- Family history – psychiatric illness, neurodevelopmental conditions, substance use.

Thorough physical examination of all systems (neurologic, cardiac, respiratory) and vital signs recorded. Information obtained through history and examination will guide the choice of investigations to be carried out. Mental status examination helps to understand a person's emotional state and cognitive capabilities and limitations. It starts with assessing

the individuals’ level of awareness and orientation to the surroundings, eye contact, communication ability, general appearance, degree of distraction, speech, affective state, thought process (worries, delusions, concerns), hallucinations, evidence of separation anxiety. Cognitive assessment includes assessing for memory disturbances, gross level of intelligence and reasoning ability – whether appropriate to age or not, understanding of problems at hand. It is always a good idea to ask for the skills/strengths/interests of the child while doing a mental status examination.

Suicidal behavior is one of the most common presenting psychiatric emergencies in children. In the assessment of suicidality, various factors need to be considered, as highlighted in [Table 2](#).<sup>[11]</sup> Key terms used have already been described in the section on suicide. (IPS CPG on assessment and management of Suicidal Behaviors).

Table 2.

Assessment of suicidality

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| Important aspects in suicidal assessment involve the following:                                                 |
| Risk to harm self and others                                                                                    |
| Clarify diagnosis                                                                                               |
| Risk factors - biological, social, and psychological risk factors                                               |
| Level of functioning                                                                                            |
| Identify strengths, support, protective factors to alleviate distress, mitigate risk of harm to self and others |
| Clarify problems and goals                                                                                      |
| Determine the most appropriate level of care for treatment                                                      |

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It is inevitable for a psychiatrist to screen all patients for suicide risk and carry out a comprehensive suicide assessment in suspected children and adolescents. Given the risk of suicide, the risk assessment is necessary to categorize risk and recommend the level of care appropriate to the child. It is also essential to ask questions in relation to suicidality as described in [Table 3](#).<sup>[12,13]</sup>

Table 3.

Important questions in interview

| Key questions                                                              |
|----------------------------------------------------------------------------|
| Have you had thoughts of wanting to harm self?                             |
| Do you have a plan as to how you would like to harm yourself?              |
| Have you ever taken steps or prepared to carry out this plan?              |
| Have you tried to harm/hurt yourself before? If so, when and how?          |
| Is there anything/anyone who would keep you from acting on these thoughts? |

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Various screening instruments are available to assess suicidality in children and adolescents.

1. ASQ (Ask Suicide Screening Questions) developed by National Institute of Mental health consists of brief suicide screening questions that take 20 s to administer.
  - i. In the past few weeks, have you wished you were dead?
  - ii. In the past few weeks, have you felt that you or your family would be better off if you were dead?
  - iii. In the past few weeks, have you been having thoughts about killing yourself?
  - iv. Have you ever tried to kill yourself? If yes, when and how?

- v. Are you having thoughts of killing yourself right now?

## Suicide risk assessment

It is a one-to-one assessment in which the distressed person is thoroughly interviewed regarding current suicidal desire/ideation, capability, intent, reasons for dying, reasons for living, suicide attempt plan, past attempts, and protective factors, thus, enabling risk identification and planning intervention. It is essential to evaluate the motivation and intent of any previous attempt and the understanding of the lethality of the suicide act or plan. The various factors to be assessed are mentioned in [Table 4](#).<sup>[14,15]</sup>

### Table 4.

#### Suicide risk in children and adolescents

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##### High risk factors for suicide in adolescents

##### Intrapersonal:

Depression (modifiable)

Alcohol and drug use (modifiable)

Previous suicide attempts (unmodifiable)

High risk behaviors (modifiable)

Sexual orientation confusion (unmodifiable)

Psychological symptoms - hopelessness, sense of losing control (modifiable)

Dysregulated sleep (modifiable)

##### Social/situation:

Stressor (modifiable)

Family factors - depression in parents, suicidal behavior, substance use disorders (modifiable)

Family violence (modifiable)

Child abuse/neglect

Lack of social support (modifiable)

Sense of isolation (modifiable)

Victim of bullying/being a bully (modifiable)

#### Others

Access to lethal means (modifiable)

Stigma associated with asking help (modifiable)

Males at a much higher risk than females (unmodifiable)

Among males - previous suicide attempters (unmodifiable)

age 16 and above (unmodifiable)

associated mood disorder (modifiable)

associated substance abuse (modifiable)

Among females - mood disorders (modifiable)

Previous suicide attempters (unmodifiable)

Immediate risk predicted by agitation and depressive disorder

Multiple suicide attempters

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The SAFE T (Suicide assessment five-step evaluation and triage):

This is a tool and a pocket card which guides psychiatrists through a comprehensive risk assessment. It includes the following five steps:[[16](#)]

1. Identify risk factors – especially noting the modifiable factors, as described in [Table 4](#).
2. Identify protective factors – especially those that can be enhanced, as described in

[Table 5.](#)

3. Make a suicide inquiry – ask questions on ideation, plan, behaviors, intent
4. Assess the risk –

The risk assessment includes

- High-risk level – severe symptoms or acute precipitating event, protective factors are not relevant, potential lethal attempt or persistent ideation with suicide intent; intervention includes – admission is indicated, suicide precautions to be administered.
  - Moderate risk level – multiple risk factors, few protective risk factors, suicidal ideation with a plan with no intent or behavior, admission may be necessary subject to risk factors. Developing a crisis plan is essential.
  - Low-risk level – Modifiable risk factors, strong protective factors present, presence of thoughts of death with no plan/intent/behavior. A patient can be managed in outpatient settings. Facilitate symptom reduction. Give emergency contact numbers.
5. Document the suicide risk level, rationale, intervention, and follow-ups.

Table 5.

Protective factors

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Protective factors:

Overall resilience

Problem-solving skills

Interpersonal and community connectedness

Safe environment

Awareness and access to physical and mental health care

Positive peer relationships

Positive adult relationships

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Protective factors: As highlighted in [Table 5](#), these are factors that mitigate/reduce suicide risk. Adolescents face challenges such as transitioning to adulthood, facing new independence, identity formation, and changing social situations at school and home. They are also at risk of anxiety or depression due to significant physical, hormonal, and social situations at school and home.[\[15\]](#)

Nonsuicidal self-injury (NSSI): Another presentation in pediatric emergencies. Defined as direct, repetitive, socially unacceptable injury to body tissues without suicidal intent. As per ICD 10 (International Statistical Classification of Diseases and related health problems), these exist as a symptom and not as a diagnostic entity. It may vary from deliberate self-harm with suicidal intent to involuntary or stereotyped behavior in developmentally disabled. A common form of deliberate self-harm is cutting and burning. It may be a way to cope with distress, punish themselves, self-soothe or manipulate the environment, or cry for help. Stereotypic behaviors include hitting, biting, head banging,

and scratching.[[14](#)] Certain risk factors which predispose to nonsuicidal self-injury include female sex, earlier NSSI, earlier suicidal ideations and attempts, symptoms of depression, and presence of stress. Also, a higher frequency of NSSI is associated with an increased risk of suicidal ideations and suicidal attempts.[[17](#)]

A clinician must be aware that NSSI and suicidal behavior are discrete entities, but they may occur in the same person at different points. [Table 6](#) shows some of the differences between suicide attempts and NSSI.[[17,18](#)]

Table 6.

Differentiate between NSSI and suicide attempt

|                   | <b>Suicide attempt</b>                                                            | <b>NSSI</b>                                                                                    |
|-------------------|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Intent            | Intent to die/lethal                                                              | No intent to die/not lethal                                                                    |
| Age               | Seen in mid adolescence                                                           | Late latency/early adolescence                                                                 |
| Gender            | Female > male (attempts)                                                          | Similar in both genders                                                                        |
| Method            | Poisoning, firearms, jumping                                                      | Scratching, banging, burning, cutting, self-hitting body parts, interfering with wound healing |
| Underlying traits | Maladaptive coping mechanism to regulate overwhelming emotions and to endure life | Desire to escape and to end life                                                               |
| Frequency         | Very few lifetime acts                                                            | Many lifetime acts                                                                             |
| Motivation        | Escape                                                                            | Temporary relief of psychic distress                                                           |
| Consequences      | Worsen depression/guilt                                                           | Relief of distress                                                                             |

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## Risk assessment for aggression

For children and adolescents presenting to the emergency department with aggressive behavior, it is important to evaluate for some of the more commonly associated psychiatric disorders with aggression like conduct disorder, oppositional defiant disorder, attention deficit hyperactivity disorder, substance use disorder, depression, disruptive behavior

disorder, autism spectrum disorder, intellectual disability, and gaming disorder.[19] There is evidence of an increase in physical aggression in adolescents with pathological gaming. [20]

A comprehensive assessment including developing therapeutic alliance, gathering detailed multi-informant history, examination, mental status examination remains the first steps to plan an effective intervention.

Performing a risk assessment is important since this involves the safety of the patient and people around. This reflects not only the safety measures that need to be undertaken but also the needs of the child and can direct intervention. Relevant factors for risk assessment are shown in Table 7.[21]

Table 7.

Risk factors and correlates for aggression and conduct disorder[22]

Biological:

- Reduced autonomic reactivity
- Decreased cortisol levels associated with proactive aggression
- HPA axis hyperactivity
- Hypo reactivity of amygdala - marker of impaired emotional processing
- Deactivation of ACC during emotional response - leading to deficient emotional processing
- Lower intelligence
- Deficits in executive functioning
- Perinatal risk factors - maternal smoking and alcohol use
- Parental psychiatric disorder
- Genetics

History of violence (especially recent)

Intoxication

Command hallucinations

Impulse control disorders

Concurrent psychosocial stressors

Verbal and physical threats

Psychomotor agitation

Paranoia

## Social

Adverse family circumstances - poverty, physical punishment, and neglect

Lack of parental discipline - inconsistent and inconsequent parenting practice (low positive involvement, more hostility, and punishment)

## Access to firearms/weapons

Psychological

Personality

Substance use

Previous history of self-harm, violence, and abuse

Adverse childhood experience

Gaming disorder

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## Assessing for reversible causes for altered sensorium

While assessing a child or adolescent who presents with behavioral abnormalities and altered sensorium additional information should be gathered with respect to the onset of symptoms, type of symptoms-altered sensorium, inattention, memory disturbances, sleep-

wake cycle disruptions, agitation, perceptual abnormalities, disturbances in thought processes, any fluctuation in symptoms, medical history including acute/chronic medical illnesses/infection, medications, pain, anemia, etc., and metabolic derangements. Altered sensorium can be due to various factors, and some of the modifiable factors (listed in the [Table 8](#)) can be remembered in the form of an acronym “BRAIN MAPS” given by Smith *et al.* 2013.[\[23\]](#)

Table 8.

Modifiable factors contributing to development of delirium in children (Smith *et al.* 2013)[[23](#)]

| Stands for | Initials of the acronym                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| B          | Bring oxygen: treat hypoxia, anemia, improve cardiac output                                                                             |
| R          | Remove/reduce drugs contributing to delirium: anticholinergics, benzodiazepines                                                         |
| A          | Atmosphere: foreign environment, bright lights, loud noises, physical restraints, frequent change in caregivers, no consistent schedule |
| I          | Infection, inflammation, immobilization                                                                                                 |
| N          | New organ dysfunction                                                                                                                   |
| M          | Metabolic disturbances: hyponatremia/hyponatremia, hypokalemia/hyperkalemia, hypoglycemia, hypocalcemia, alkalosis, acidosis            |
| A          | Awake: disturbances of sleep wake cycle, lack of consistent bed time routine                                                            |
| P          | Pain: too much pain, undertreated pain, overtreated pain                                                                                |
| S          | Sedation: assess the need for sedation and appropriate sedation                                                                         |

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## Assessment for causes leading to the presentation as psychosis

Assessment of children presenting with psychosis requires knowledge of various causes that can present with psychosis as a presenting complaint, and some of them are listed in

Table 9.

Causes of psychosis/psychotic symptoms in children

| Specific disorders                                                      | Etiological factors                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Known CAUSES                                                            |                                                                                                                                                                                                                                                                                                                                                      |
| Central nervous system                                                  |                                                                                                                                                                                                                                                                                                                                                      |
| Rheumatological conditions Toxins                                       | Infections: Herpes simplex encephalitis, arboviruses, measles encephalitis, subacute sclerosing                                                                                                                                                                                                                                                      |
| Nutritional deficiency Inborn errors of metabolism Autoimmune disorders | panencephalitis, HIV, Epstein bar virus, meningitis, tuberculosis, cerebral malaria, toxoplasmosis                                                                                                                                                                                                                                                   |
| Others                                                                  | Neurodegenerative disorders: multiple sclerosis, Huntington’s chorea                                                                                                                                                                                                                                                                                 |
|                                                                         | Epilepsy: temporal lobe epilepsy, postictal psychosis, Landau-Kleffner syndrome                                                                                                                                                                                                                                                                      |
|                                                                         | Head injury Stroke CNS mass lesions: tumors, abscess Hydrocephalous                                                                                                                                                                                                                                                                                  |
|                                                                         | Vascular: venous thrombosis, ischemia, aneurysm Systemic lupus erythematosus, sarcoidosis Lead poisoning, carbon monoxide poisoning, organophosphate poisoning Anemia, vitamin B12 deficiency, vitamin D deficiency Adrenoleukodystrophy, lysosomal disorders, cerebrotendinous xanthomatosis, homocystinuria, urea cycle defects NMDA encephalitis, |

|                                |                                                                                                                                                                                                                                                  |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                | Hashimoto's encephalopathy, thyroid storm, antiphospholipid syndrome<br>Wilson's disease, acute intermittent porphyria                                                                                                                           |
| Substance overdose/abuse       | Volatile substances, hallucinogens (lysergic acid, phencyclidine), marijuana, datura, MDMA, amphetamines, methamphetamine, cocaine, bath salts                                                                                                   |
| Prescription drug side effects | Anticholinergics, decongestants (pseudoephedrine), steroids, isoniazid, antibiotics (amoxicillin, clarithromycin, erythromycin), antiepileptic drugs (phenytoin, topiramate, levetiracetam), statins, antiviral agents, immunosuppressive agents |
| Drug related syndromes         | Serotonin syndrome, neuroleptic malignant syndrome, baclofen withdrawal, benzodiazepine withdrawal, sudden psychotropic withdrawal                                                                                                               |
| Primary psychiatric illness    | Depression, bipolar disorder, early onset schizophrenia, acute and transient psychosis Trauma and abuse, emotional issues                                                                                                                        |

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For assessment of children presenting with psychotic symptoms, a thorough medical evaluation should be carried out and medical stabilization should be done whenever required. Drug ingestion/overdose whether intentional or unintentional may not be recognizable immediately as children are not usually in a position to provide information.

So, efforts for rapport building with the child and the family should start at the first contact and continue thereafter. Additionally, it is imperative to provide support to the caregivers when their child is being evaluated. *It is also to be remembered that transient psychotic-like symptoms may be just a developmental phase in the young child; however, such presentations require a careful information elicitation and continued monitoring of symptoms.* [\[10\]](#)

Individuals with a primary psychiatric illness usually have normal vital signs; normal orientation to surroundings and hallucinations are mostly auditory in nature. In contrast, individuals with psychotic symptoms in the context of an underlying medical condition may have abnormal vital signs, disorientation, positive signs on physical examination, visual, and tactile hallucinations. There are certain indicators for an underlying medical condition in the context of psychotic symptoms, which can be helpful in arriving at a differential diagnosis. These are:

- New and recent onset symptoms and/or behavioral change
- Younger children
- History of substance use/recent medication use
- Abnormal vital signs, abnormality of physical examination and/or neurological examination
- Symptoms like disorientation, visual hallucinations, fluctuations in mental status, emotional lability, etc.

One should always clinically assess for the altered sensorium since altered sensorium and psychosis in young children can be difficult to differentiate.

## Assessment of anxiety and agitation in children and adolescents

Anxiety and agitation can have a myriad reason. One needs to differentiate from developmentally appropriate worries, fears, and responses to a stressor, as described in [Table 10](#). [\[24,25\]](#)

Table 10.

Difference between developmentally normal anxiety and pathological anxiety

|            | <b>Developmentally normal anxiety</b>                                                       | <b>Pathological anxiety</b>                                                                        |
|------------|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Intensity  | As per developmental age and event, the anxiety is realistic                                | The degree of anxiety is unrealistic as per developmental stage and event                          |
| Impairment | No interference in daily life<br>Dysfunction seen in academics, friendships and family life | Impairment in academic and family life                                                             |
| Course     | Usually remit                                                                               | Chronic and persistent and linked with poor long-term functioning, suicidality, and general health |
| Treatment  | Usually these children experience remission                                                 | Pharmacological and nonpharmacological approaches are used                                         |

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Detailed clinical history and examination help ascertain key areas of concern and presence (or absence) of problems. A patient may present with an overwhelming surge of anxiety, stress, and fear. They may present as distress, cry spells, tantrums, freezing, clinging, or not wanting to leave a familiar person. Panic attacks peak in late adolescence, affecting 5–10% of adolescence. They express a fear of death with autonomic symptoms such as tachycardia, palpitations, sweating, shortness of breath, chest pain, choking sensation, nausea, abdominal pain, tremors, tingling, and numbness. Children with marked stress and fear in social situations may cry and throw a tantrum and present as a panic attack.

Children who experience trauma following an actual or threatened death, accident, injury, or threat to physical integrity or witness an event (such as sexual abuse, assault, shooting, or an earthquake) may present with PTSD. A response in the form of intense fear, helplessness, and horror may present in emergency settings.[24] In an emergency setting, altered consciousness with fluctuating attention needs to be differentiated from trance states or black outs. They may also have depersonalization or derealization episodes.[26] A child's belief in an alternate self or imaginary self which control child's behavior and may present in the ED.

Comorbid psychiatric illnesses which may have anxiety include (but are not limited to) depression, attention deficit hyperactivity disorder (ADHD), and behavior, bipolar, obsessive-compulsive, eating, learning, language, and substance-related disorders. It is necessary to look for comorbid medical illness, hyperthyroidism, caffeinism, migraine, asthma, diabetes, chronic pain/illness, lead intoxication, hypoglycemic episodes, hypoxia, pheochromocytoma, central nervous system disorders, cardiac arrhythmias, cardiac valvular disease, systemic lupus erythematosus, allergic reactions, and dysmenorrhea. Medications that can cause anxiety include (but are not limited to) bronchodilators, nasal decongestants and other sympathomimetics, antihistamines, steroids, dietary supplements, stimulants, antidepressants, antipsychotics, and withdrawal from benzodiazepines (particularly short-acting).[27] In children who present with seizure-like episode, a clinician needs to differentiate between a dissociative convulsion and an epileptic seizure.[28]

## USE OF RATING SCALES

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There are various rating scales that can be used to augment information obtained by clinical interviews. Some of the rating scales that can be used for various presenting emergencies are listed in [Table 11](#).

Table 11.

Rating scales used in children and adolescents in the emergency setting

| Scale                                                           | Validated age of use          | Administered by and use                                 | Remarks                                                                            |
|-----------------------------------------------------------------|-------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------|
| For assessment of suicidal behavior                             |                               |                                                         |                                                                                    |
| Self-completion by child and adolescent                         |                               |                                                         |                                                                                    |
| Beck hopelessness scale (BHS)[ <a href="#">29</a> ]             | Adolescents                   | Assess hopelessness                                     | 17 True and false items, clinical/research and screening                           |
| Columbia Teen Screen (CTS)[ <a href="#">29</a> ]                | Adolescents (11-18 years old) | Screen for suicidal behavior, ideation and risk factors | 26 Item clinical/research and screening. High sensitivity and specificity          |
| Suicidal Ideation Questionnaire (SIQ) [ <a href="#">30</a> ]    | Adolescents (11-18 years old) | Measures frequency and severity                         | Research and screening High sensitivity and specificity                            |
| Suicide probability Scale[ <a href="#">31</a> ]                 | Ages+14 years                 | Clinical index of suicide risk                          | Clinical purpose Self rated                                                        |
| Child Adolescent Suicidal Potential Index[ <a href="#">14</a> ] | 6-17 years old                | Assess suicidal behavior                                | 30 yes/no items clinical/research and screening Excellent reliability and validity |
| Clinician administered: child and adolescent                    |                               |                                                         |                                                                                    |
| Child suicide potential scale[ <a href="#">14</a> ]             | 6-12 years old                | Assess suicidal behaviors and risk factors              | Clinical and research use                                                          |
| Suicide potential                                               | 11-18 years                   | Suicide risk                                            | Diagnostic, research,                                                              |

|                                                                          |                                         |                                                                                                                                           |                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| interview[ <a href="#">14</a> ]                                          | old                                     | assessment                                                                                                                                | and screening                                                                                                                                                                                                                                                                                      |
| Columbia Suicide severity Rating scale (C-SSRS)[ <a href="#">14,32</a> ] | Has been tried children <5 years of age | Assess the patient's responses to screening questions. Validated in emergency situations too                                              | Translated in more than 30 languages. Has been used as a screening tool for suicidality                                                                                                                                                                                                            |
| For assessment of aggression                                             |                                         |                                                                                                                                           |                                                                                                                                                                                                                                                                                                    |
| Child behavior checklist (CBCL) [ <a href="#">33,34</a> ]                | Ages 6-18 years                         | Parent report of ADHD, ODD, and CD.                                                                                                       | 112 items rated on three-point scale, plus social activity/academic performance subscales Yields internalizing scores (subscales: withdrawn, somatic complaints, anxious/depressed) and externalizing scores (subscales: Delinquent behavior and aggressive behavior) Screening and tracks outcome |
| Modified Overt aggression scale (MOAS)[ <a href="#">35</a> ]             |                                         | Rates behavior over a 1 week. Verbal Aggression against property Autosuggestion Physical Five -point response format Allows assessment of | For outpatient settings                                                                                                                                                                                                                                                                            |

Impulsive/Premeditated  
Aggression Scale (IPAS)  
[36]

both severity and  
frequency

Self-report. Half of  
questionnaire items  
correlate with  
impulsive  
aggression, whereas  
the other correlates  
with pre meditated  
aggression.

Children's Aggression 5-18 years  
Scale - Parent and old  
Teacher versions (CAS)  
[37,38]

Frequency and  
severity of  
aggression in  
children and  
adolescents

Five domains: Verbal  
aggression,  
aggression against  
objects and animals,  
provoked physical  
aggression,  
unprovoked physical  
aggression, and use  
of weapons  
Distinguishes  
aggression 1) inside  
vs. outside the home  
and 2) against  
children vs. adults

For assessment of  
altered sensorium

Pediatric Anesthesia > 1 year  
Emergence Delirium  
Scale (PAED)[39]

Clinician rating on  
behavior  
observations: Eye  
contact, goal-  
directed outcome,  
awareness of  
surroundings,

Bedside rating scale  
to detect emergence  
or hyperactive  
delirium; not useful  
to detect hypoactive  
delirium

restlessness, whether  
inconsolable child  
Rating from 1 (not  
at all) to 5  
(extremely)

Cornell Assessment  
of Pediatric delirium  
(CAPD)[[40-42](#)]

Children of  
all ages

Caregiver rated,  
eight-item scale.  
Scoring from 0 (not  
at all) to 4  
(extremely). Scores  
≥9 indicates  
delirium

Bedside tool, easy to  
administer by  
caregiver, even in  
critically ill children,  
can pick subtle  
behaviors over time,  
useful to identify  
hypoactive and  
hyperactive delirium

pCAM-ICU psCAM-  
ICU[[43-45](#)]

>5 years 6  
months-5  
years

Clinician rated;  
based on DSM-IV-TR  
criteria. Acute  
change or  
fluctuation in mental  
status Inattention  
Altered level of  
consciousness  
Disorganized  
thinking If (1) nor  
(2) is present -  
negative screen for  
delirium If (1) and  
(2) are present plus  
either (3) or (4)  
present - positive  
screen for delirium

Can also be used on  
children receiving  
mechanical  
ventilation. It  
follows a two-step  
process for  
assessment of  
delirium: 1) Arousal  
assessed by a  
sedation scale 2)  
Delirium is assessed  
if patient is at least  
arousable to voice

Sophia Observation  
Withdrawal Symptoms-  
Pediatric Delirium Scale

3 months-  
16 years

Clinician rated scale.  
22 items-17 items  
pertaining to

Easily administered  
in 2-5 min

(SOS-PD)[[46](#)]

symptoms of  
pediatric delirium  
(PD) and 15 items  
pertaining to  
symptoms of  
iatrogenic  
withdrawal  
syndrome (IWS)  
resulting from  
prolonged  
administration  
and/or high doses of  
benzodiazepines or  
opioids; 10  
overlapping items.  
“Yes/No” response  
for symptoms  
observed in the  
previous 4 h

For assessment of  
psychosis

Brief psychiatric  
Rating Scale for  
children (BPRS-C)[[47](#)]

5-18 years  
of age

Assess emotional  
and behavioral  
problems in  
children; 21 items  
scored on a seven-  
point Likert scale

Easy to administer;  
takes 5 min

For assessment of  
anxiety

Pediatric Symptom  
Checklist[[48](#)]

primary  
care,  
school, or  
other child-

It is  
social–emotional  
screening  
instruments

serving  
settings

The  
Multidimensional  
Anxiety Scale for  
Children[[49](#)]

>8 years

The Screen for Child  
Anxiety and Related  
Emotional Disorders  
(SCARED)[[50](#)]

>8 years

The Spence  
Children's Anxiety Scale  
(SCAS)[[50](#)]

>8 years

Preschool Anxiety  
Scale[[50](#)]

2.5-6.5  
years

Parent report  
adapted from the  
SCAS that was  
developed for  
screening for anxiety  
in young children

The Social Anxiety  
Scale, the Social  
Worries Questionnaire,  
and the social phobia  
subscale of  
SCARED[[51](#)]

>8 years

Brief screening  
measures for social  
phobia/social  
anxiety symptoms

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## INVESTIGATIONS

The choice of special investigation to be advised (e.g., MRI brain, lumbar puncture, EEG,

autoimmune panel, urine for porphobilinogens, specific metabolic studies, etc.) usually is guided by the information gathered through history and examination. The investigations may include a complete blood count, toxicology screen for substances, renal function test, liver function test, electrocardiogram, neuroimaging, and many more.

## DIAGNOSIS

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Diagnosis should be made as per the standard nosological system. However, it should be remembered that a final diagnosis may not be easy to make in an emergency setting (underlying medical conditions will be picked up on investigations), and some psychiatric symptoms like aggression, suicidal behavior, agitation, altered behavior, psychotic symptoms are a presentation of many disorders. Therefore, a thorough history, physical and neurological examination, and mental status examination will help to come to a provisional diagnosis.

## MANAGEMENT OF PSYCHIATRIC EMERGENCIES IN CHILDREN AND ADOLESCENTS

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### The decision for admission

The main goal of assessment of children and adolescents in the emergency setting includes triage and focused care, medical stabilization, ensuring the safety of the child or adolescent, relevant investigations, diagnostic clarification, deciding the treatment setting (inpatient- psychiatry, inpatient-pediatrics, intensive care unit or outpatient services). Patients with severe symptoms often warrant admission.

Medical stabilization includes care of hemodynamic status (A – airway, B – breathing, C- circulation), nutritional status, physical activity, pain, sedation. Thereafter, the first step should be an assessment for altered behavior/delirium. In any case where altered behavior is present, the management of altered behavior takes precedence (the management of altered behavior is described in the sections below).

Children and adolescents with active suicidal ideations and attempts should be

hospitalized if the child's condition is unstable. Caregivers' supervision is to ensure that drugs and any means of self-harm should be inaccessible to the child/adolescent at risk. [\[14\]](#)

Steps for suicide prevention in wards are necessary. It is essential to remodel wards to prevent suicide attempts in the ward. Eliminate structures that support hanging objects, exposed pipes, and towel hooks. Install windows that do not open from the inside. Remove harmful objects from the vicinity. Nonsharp utensils are advised to be used. Attendant or caregiver supervision is to be maintained throughout. In children and adolescents, partial hospitalization offers intensive multidisciplinary treatments and skilled observation and support. This can be offered in those who are disturbed but have a supportive care environment. This allows intervention to stabilize the emotional condition and address stressors.

## Treatment in emergency services

The treatment will vary depending on the cluster of symptoms.

### Self harm

A detailed discussion with the family and child about specific issues, triggers, and situations that result in suicidal behavior must be carried out. A written, verbal no-suicide contract is commonly discussed with the child and is also used as a probe to understand if the thoughts on suicide change in the child. A verbal or written no-suicide contract has not shown evidence of preventing subsequent suicides. [\[14\]](#) However, safety planning should be conducted in the emergency department.

It is here that the mental health professional provides an important function of triage, referring suitable patients for subsequent treatment. Rotheram-Borus *et al.*, 1996 [\[52\]](#) described a brief emergency room crisis intervention procedure for adolescent attempters. The aims are to facilitate a good experience between family and emergency service staff, set realistic expectations about follow-up treatment, and obtain a commitment from adolescents and relative toward follow-up. A detailed discussion with the patient and

family about specific issues or situations which enable suicidal behavior, ongoing stressors, and identifying potential precipitating factors are important.[14] In the emergency situation, once the wound care and dressing are completed, a complete psychological assessment is necessary. Upon evaluation when it is clear that the self-harm is NSSI, then interventions focusing on developing motivation to change, facilitate family support, and strengthen positive affect can be taken up in regular outpatient care.[53]

## Safety planning

A safety plan is a document with six steps where the clinician and patient discuss warning signs, coping strategies, ways to reach out for help, and make an environment safer. It involves collaborative work in which the individual develops a personalized list of coping strategies. The steps of safety planning are shown in Table 12.[11,12]

Table 12.

### Safety planning measures

|                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Safety planning                                                                                                                                         |
| Discuss warning signs (mood, behavior, thoughts, images, situation)                                                                                     |
| Internal coping strategies (relaxation exercise, physical activity) - things I can do to take my mind off my problems without contacting another person |
| People and social settings that provide distraction People whom I can ask for help                                                                      |
| Professionals or agencies I can contact during a crisis (Lifeline)                                                                                      |

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### Safety plan

## Example of elements of a safety plan

### Step 1: Warning sign

- I feel hopeless and suicidal
- Witnessed an argument between my parents
- Thoughts of previous suicidal attempt

### Step 2: Internal coping strategies – Things I can do to distract myself/cope

- Listen to music
- Read a book
- Journal my thoughts

### Step 3: Social situations or people that can help to distract me

- Talk to mom
- Talk to my cousin

### Step 4: People whom I can reach out to for help

- Ask Dr ABC for help in .... Hospital
- My teacher PQR mobile:

### Step 5: Professionals or agencies I can contact during a crisis:

1. Dr. Name/mobile no./emergency contact
2. School counselor Name/mobile no./emergency contact
3. Local hospital emergency
4. Suicide helpline

Make the environment safe:

1. Medications with mom
2. No harmful objects in the vicinity.

A study done on adolescents to assess the association between profiles and mental health utilization reveals five profiles of elevated suicide risk with differing patterns of risk factors which includes the history of multiple suicide attempts, suicidal ideations in the last month, depression, substance use, aggressive behavior, and abuse.[\[54\]](#)

## Pharmacologic treatment

Pharmacologic approaches involve the treatment of the underlying psychiatric condition.

Lithium reduces suicidality and suicide attempts in previous attempters. However, in children and adolescents' supervision is to be exercised in view of a potential overdose in suicidal children.[\[14\]](#) SSRIs (selective serotonin reuptake inhibitors) are used to treat the underlying depression. These are considered first-choice medications in suicidal children and adolescents. The clinician must keep a watch on emerging side effects such as suicidal ideas in such patients. One of the major concerns with the use of antidepressants among children and adolescents is the emergence of suicidal ideations (Black box warning) against the use of antidepressants. A cautious approach needs to be considered using antidepressants among children and adolescents, and they must be monitored for any treatment for emergent suicidal behavior. Clozapine can also be used in children and adolescents with suicidality as off label use.[\[14\]](#) Studies have reported the use of ketamine in youth with a decrease in suicidality; however, findings need to be substantiated with longitudinal studies, safety, efficacy, and abuse potential in youth.[\[55\]](#)

## Specific psychotherapies

Whilst crisis intervention is possible in emergency settings, once stabilized the child may be engaged in other forms of therapy, such as cognitive behavioral therapy (CBT), interpersonal psychotherapy, and dialectic behavior therapy. Family therapy is specifically designed to treat depression and suicidal thoughts/behaviors. It is to protect adolescents against suicidal ideations and risk behaviors by improving family processes and securing

parent–child bond.

When children and adolescents report suicidal ideas, they experience intolerable agony, hopelessness, and helplessness. They may impulsively respond to desperation by attempting suicide. Psychotherapeutic techniques aim at reducing intolerable feelings and thoughts and reorienting the cognitive and emotional perspectives of the suicidal child and adolescent.[\[14\]](#)

It is essential for the clinician to relate to the child

- Honest and consistent way
- Understand suicidal patient's attitude and life problems
- Convey and instill hope and optimism.

Electroconvulsive therapy: Suicidality is one indication for electroconvulsive therapy in children. ECT is considered in patients with a failure to respond to two adequate trials or in conditions that are life-saving. As per the Mental Healthcare Act, 2017, Electroconvulsive therapy can be given to children with informed consent obtained from guardians and with permission from the Mental Health review board.[\[56\]](#) However, this cannot be in an emergency setting.

## Follow-up and discharge planning

A checklist can be kept in mind before discharging a patient who has attempted suicide.

Children and adolescents shall never be discharged from the emergency without the verifying account from the parent.

- Ensuring supervision by a supportive person at home
- Inaccessibility to lethal means (drugs, weapons, knives, harmful objects)
- Drug dose/duration
- Check that follow-ups are closely spaced.[\[14\]](#)

The doctor and parent need to be advised to monitor for warning signs:

- Thoughts of dying – if the child voices thought of dying, disappearing, shooting, other forms of self-harm, or jumping.
- Change in behavior – sad, withdrawn, irritable, anxious, restless, indecisive, difficulty in concentration
- Change in sleep patterns – early awakenings, excess sedation, nightmares, insomnia
- Change in eating habits – loss of taste and appetite, overeating.

## Aggression

The Centre for Education and Research on Mental health therapeutics (CERT) guidelines for the Treatment of Maladaptive Aggression (TMAY) II recommend the following:[\[57\]](#)

1. Intervene by giving evidence-based psychotherapy
2. Engage the child, family, and school in psychosocial strategies
3. Initiate psychopharmacologic treatment for psychiatric conditions
4. Evidence-based guidelines to treat primary (underlying) disorders
5. Residual aggression persists and then treat with atypical antipsychotic.

Treatment planning in an aggressive patient should include a review of aggressive behavior, including triggers, warning signs, repetitive behaviors, response to treatment, and prior seclusion and restraint events associated with aggressive acts, along with diagnosing and treating the underlying psychiatric illness. Cognitive limitations, neurological deficits, and learning disabilities need to be taken into consideration.

De-escalation strategies include helping patients manage anger outbursts by using anger management and stress reduction techniques. The de-escalation strategies include allowing children to use self-direction, prompts to manage their own behavior, ignoring peer provocation, negotiating with peers, and use self-directed time out. Prompts that aid children in using these strategies should be a primary treatment intervention and should be included in the treatment plan.

Aggression is often preceded by a period of escalating agitation. Psychosocial interventions help regain self-control and can help avoid the need for physical and chemical restraints. When the youth exhibit warning signs of aggression, behavioral interventions such as verbal interventions, time out, or quiet time can be used. A stat dosing or injectable can be used to avoid the aggressive episode when anticipated. In case less restrictive options fail and there is a danger to self or other and/or imminent danger of damaging property, restraint can be used. All restrained patients should have their pulse, blood pressure, and range of motion in their extremities checked every 15 min. (Refer to the section on aggression in psychosis.)

As per the Centers for Medicare and Medicaid Services restraint guidelines, regulations apply to both physical and chemical restraints. Documentation is required for the need and monitoring of restraint of 100% of patients.

- Under 9 years of age – every 1-h restraint is to be renewed
- From 9 to 17 years of age – every 2-h restraint is to be renewed
- Above 18 years of age – every 4-h restraint is to be renewed. The outline of the assessment and management of self harm can be found in [Algorithm 1](#).

## Monitoring and basic care requirement

- Visual check – every 15 min or constant observation
- Release a restraint – every 2 h (may reapply if needed)
- Neurovascular check – every 2 h
- Offer food/water/bathroom – every 2 h
- Behavior check – every 2 h
- Respiratory status check – every 2 h
- Change physical position – every 2 h.

## Verbal restraint

The clinician can introduce oneself, prepare the patient for what will happen, and respect the patient's autonomy. It is imperative to also offer food and liquids and offer empathic

listening. A clinician must keep in mind asking about patient's requests/preferences and honoring reasonable requests. Simple, direct, and soft language with a decrease in stimulation is important. Reassure the patient that they will be safe and offer distraction. Ensure to remove brittle objects/equipment in the vicinity of the patients.

## Physical restraints

Supine position, avoid covering the patient's face/mouth/nose. Avoid pressure on the neck/back and chest. Elevate the head of the bed, if possible. [\[58\]](#)

## Pharmacological approaches

Taking into account the child's age and development, in addition to psychotherapy, pharmacological approaches are used in patients with aggression. In very young children and preschoolers, caution is to be exercised and nonpharmacologic approaches are preferred. It is important to "start low and go slow" and regularly assess for side effects and progress.

Stimulants like methylphenidate are efficacious in reducing aggression in children with ADHD, and disruptive behavioral disorder and improvement is also seen in oppositional behavior and aggression. Alpha agonists have weak evidence in improving aggression in children and adolescents. SSRIs have shown efficacy in improvement in irritability in children and adolescents with severe mood dysregulation, whereas atypical antipsychotics have strong evidence for use and efficacy in the treatment of aggression in different psychiatric disorders.

Among antipsychotics, risperidone is used as the first-line treatment for aggression, especially in autism spectrum disorder, ADHD, and conduct disorder. Aripiprazole is also used for the treatment of irritability associated with autism spectrum disorder, and there is off-label use of aripiprazole for aggression. Other atypical antipsychotics such as olanzapine, clozapine, quetiapine, ziprasidone, lurasidone have shown some evidence in reducing aggression. Treatment recommendations for the Use of Antipsychotics for Aggressive Youth (TRAAY) recommend tapering of antipsychotics after 6–9 month period

without aggression. The outline of the assessment and management of aggression can be found in [Algorithm 2](#). Typical antipsychotics are usually not recommended due to the risk of tardive dyskinesias. It is important to monitor the safety of antipsychotics medications by educating parents and children. Valproic acid can be used in patients with aggression in disruptive behavior disorder and mood lability. Lithium reports lower aggression in patients with temper outbursts, physical aggression, and explosiveness in youth. Various drugs with doses have been described in [Table 13](#).[\[59\]](#)

Table 13.

## Drugs and aggression

| Drugs        | Initial dose                                                           | Onset                              | Half-life (hours) |
|--------------|------------------------------------------------------------------------|------------------------------------|-------------------|
| Lorazepam    | 0.05-0.1 mg/kg Teen - 2-4 mg                                           | 20-30 min (PO)<br>5-15 min (IM)    | 12                |
| Midazolam    | 0.05-0.15 mg/kg Teen - 2-4 mg                                          | 20-30 min (PO)<br>5-15 min (IM)    | 3-4               |
| Haloperidol  | mg/kg Teen - 2-4 mg                                                    | 30-60 min (PO)<br>15-30 min (IM)   | 21                |
| Risperidone  | <12 yrs - 0.5 mg Teen - 1 mg                                           | 45-60 min (PO)                     | 20                |
| Olanzapine   | <12 yrs - 2.5 mg Teen - 5-10 mg                                        | 45-60 min (PO)<br>30-60 min (IM)   | 30                |
| Aripiprazole | <12 yrs - 1-2 mg Teen - 2-5 mg                                         | 60-180 min (PO)<br>30-120 min (IM) | 75                |
| Clozapine    | 6.25 mg starting dose 150-300 mg in children 200-600 mg in adolescents | 1.1-3.6 h                          | 5-16 h            |
| Ziprasidone  | <12 yrs - 5 mg Teen - 10-20 mg                                         | 60 min (PO) 30-60 min (IM)         | 2-7 h             |
| Quetiapine   | Child - 12.5 mg Adolescent - 25-50 mg                                  | 1-2 h                              | 6 h               |

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## Psychotherapy for aggression

While crisis intervention may have some role, a therapy is offered in nonemergency

settings. Therapies for aggression are based on social learning theory and developmental principles taking into account the age of the child.

In younger children, effective programs are behavior modification and emphasis on helping parents improve parent–child interactions. In children less than 12 years or younger, multiple evidence-based parent-centered programs have been effective. These deal with parenting skills, enhancing the relationship between parent and child through effective emotional communication and parent management training. Child-centered programs are used where elements of CBT focus on assisting the child in identifying triggers of aggression, challenge cognitive perceptions and improved problem-solving skills. Family-centered programs address dysfunctional interactions between family members, focus on prosocial skills such as cognitive reframing, attentive listening, and address unresolved conflicts that perpetuate aggression. Multimodal psychotherapies such as multisystem therapy and others combine different methods of intervention addressing the needs of the child.[\[21\]](#) In patients with gaming disorder, CBT and family interventions are beneficial.[\[60\]](#)

## Delirium

There is very little literature available on the management of delirium in children, and thus, clinical practice is based on expert opinion or on information extrapolated for literature on delirium in adults.[\[61\]](#) It is generally recommended that the three-pronged approach for the management of adult delirium can be readily applied to the management of delirium in children. It involves the identification and management of predisposing factors, which gives a clue to the underlying etiology of delirium. The three most common causes of delirium in children are infections, medication-related factors, and autoimmune-related factors.[\[62\]](#) There should be judicious use of the most common deliriogenic medications, namely anticholinergic agents (diphenhydramine, atropine, scopolamine, chlorpheniramine, hyoscyamine), benzodiazepines, and opioids. The table enlisting the various modifiable factors (BRAIN MAPS) can be used here. This will also direct the investigations that need to be done. There are no FDA-approved drugs for the prevention and treatment of delirium in children. Pharmacologic treatment may be required when the symptoms of delirium interfere with the care and safety of the child and the underlying cause is either under evaluation or one that cannot be immediately remedied. Additional

benefit of pharmacological treatment of delirium in children is to shorten the duration of delirious state, reduce emotional trauma and distress associated with invasive procedures and hospitalization, and to shorten hospital stay. Antipsychotics are the preferred drugs, always starting on the lowest dose and then titrating depending upon the child's response. There is no recommended dose range established for use in delirium in children; however, it is advised to start low and go slow. Among the antipsychotics, haloperidol is especially useful parenterally when oral medications cannot be given/not tolerated/severe agitation is present.[63] Second-generation antipsychotics like risperidone, olanzapine, and quetiapine are also useful in the treatment of delirium. The emergence of side effects like tachycardia, QTc prolongation, hypotension, sedation, and extrapyramidal side effects should be monitored. Nonpharmacological management of delirium is highlighted in [Table 14](#). [6,64] Although melatonin is being used for the management of delirium in adults, its use in the pediatric population is less researched. A retrospective single-center study in children aged 1–18 years admitted to PICU (pediatric intensive care unit) reports that the use of melatonin for delirium did not decrease the use of antipsychotics for the management of delirium. Hence, use of melatonin in managing pediatric delirium requires further research.[65]

Table 14.

Nonpharmacological management of delirium

| Comments                     | Intervention                                                                                                                                                                                                                                                                                                                        |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Environmental modification   | Calm, reassuring, consistent, adequately lighted, and predictable environment Pictures of family, home, etc., can be put in the room Children who wear glasses or hearing aids should wear them when possible Avoid frequent change of care givers                                                                                  |
| Promotion of sleep           | Optimization of noise and light exposure to promote sleep. Sleep during day except scheduled rest times should be discouraged Eye mask to block light during sleep Avoid overstimulation particularly before bed time                                                                                                               |
| Providing reorientation cues | Calendar and clock for day and time identification Picture of sun during day and moon during night Actively reorient the child                                                                                                                                                                                                      |
| Communication                | Speak in a calm and slow pace using clear and short sentences Explain why and where the child is Do not make continued attempts to engage the child; do not argue with the child for any perceptual abnormalities the child might be reporting Do not put questions to the child; rather remind the child about the time of the day |

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Caregiver involvement in the management of delirium in children: management of delirium in children is incomplete without addressing the carers' anxieties and dilemmas. Providing frequent reassurance and guidance about the behavior of the child, fluctuations in symptoms, their role in nonpharmacological management of delirium, and the expected

outcome of the condition are essential. [\[63\]](#)

## Psychosis

In cases presenting with psychotic symptoms in the absence of altered sensorium/delirium, the initial steps should be directed toward trying to identify the organic causes for psychotic symptoms. Thus, information obtained along with detailed general physical examination and neurological examination gives clues for relevant investigations to be done. In cases with focal neurological deficits, relevant investigations and appropriate care and referral should be done. In cases where no focal neurological deficit is evident, thorough hematological parameters, i.e., complete hemogram, liver function test, renal function test, thyroid function test, serum ammonia level, and toxicology screen should be done. In cases with preserved sensorium and no abnormality in investigations, one should consider a pediatric neurology consultation and plan further investigations to evaluate for autoimmune encephalitis, porphyrias, Wilson's disease. Where abnormality in investigations is detected, appropriate referral and management should ensue, and in cases where no abnormality in investigations is detected, the case should be managed as per the guidelines for the management of primary psychiatric illness.

Pharmacological management involves the treatment of an underlying medical condition in cases where psychotic symptoms are present in the context of a medical illness, along with treatment of agitation and violent behavior as the situation demands. However, a step-wise approach should be followed in the initial management of a violent and/or agitated child.

- Verbal de-escalation: The aim of verbal de-escalation is to distract the child by talking to him/her.
- Show of force: It is another calming method that elicits cooperation.
- Chemical restraint: Sedative/hypnotics are used preferably orally. Lorazepam (0.05–0.1 mg/kg) or haloperidol (0.025–0.085 mg/kg; maximum 5 mg) orally is an alternative to benzodiazepine sedation in an acutely psychotic child or adolescent. [\[66\]](#) Due to high propensity of side effects due to the use of typical antipsychotics, atypical antipsychotics (olanzapine, risperidone, quetiapine) can also be used.

- Physical restraint: “Therapeutic holding” is a less severe form of physical restraint where the child/adolescent is held by two persons to allow for time to the child to regain control over his/her behavior. This method is less confining and is allowed up to a maximum of 10 min and should be used prior to physical restraint, in cases where physical restraint is unavoidable and is the last resort available. Proper and adequate documentation about the nature of emergency, patient benefit, and parental consent is mandatory. It should be used for the minimum possible time with continuous monitoring of vital signs.

Pharmacological management includes treatment with antipsychotics, antidepressants, mood stabilizers, and benzodiazepines, depending upon the primary psychiatric diagnosis. [67] The outline of diagnostic approach with a child with altered behavior and psychotic symptoms can be found in [Algorithm 3](#).

## Anxiety

Good therapeutic rapport and alliance with the child are must. It is also very important to have a good doctor–parent relationship because the parents have to become an ally in the treatment of the child. Parents should be involved. In the process of the assessment of the child, the family should be strongly assured that there is nothing physically seriously wrong with the child and the symptoms are psychogenic.[28]

Acute treatment[28]: This includes the important aspects of care as highlighted in [Table 15](#). The outline of management with anxiety can be found in [Algorithm 4](#).

Psychoeducation also forms another important aspect of management, as highlighted in [Table 16](#).

## Table 15.

Acute treatment includes the following

---

Assurance  
Relaxation/de escalation  
Doctor-child and family relationship  
Restoration of communication in the family  
Realistic solution of the problems  
Reduction of sick role and secondary gains  
Promotion of positive behavior  
Teaching healthy coping  
Treatment of comorbid psychiatric or physical disorders  
Treatment of chronic condition on follow up  
Family-focused cognitive behavior therapy

---

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## Table 16.

### Important points for psychoeducation[28]

---

It should be acknowledged that the child has real symptoms and sufferings, but the reasons could be psychological and not physical

Harmlessness of the symptoms should be emphasized. It should be emphasized that symptoms are not dangerous or fatal symptoms in the absence of serious physical disease are common

These symptoms could be better explained by the mind-body relationship, for example, anxiety can lead to palpitations, tremors, rapid breathing, sweating, etc.

Emotions can cause physical symptoms and this can happen even in children

Stressors may not be severe or unimportant from an adult point of view but may be very important matter of concern for the child

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### In an emergency setting

- Crisis intervention
- Supportive therapy is implemented

CBT and mindfulness-based therapies can be considered for follow-up.[24]

Pharmacotherapy: Whilst de-escalation and reassurance are the first steps, in severe cases, pharmacotherapy plays a role.[68] [Table 17](#) shows the various drugs along with doses and indications for the treatment of anxiety.

Table 17.

Treatment of anxiety - drugs, doses, and indications

|                                         | Dose                                       | Age  |
|-----------------------------------------|--------------------------------------------|------|
| SSRIs                                   |                                            |      |
| Selective serotonin reuptake inhibitors |                                            |      |
| Fluoxetine                              | 20 mg/d                                    | 7-17 |
| Fluvoxamine                             | 50-250 mg/d child, max 300 mg/d adolescent | 6-17 |
| Sertraline                              | 50 mg/d                                    | 5-17 |
| Paroxetine                              | 10-50 mg/d                                 | 8-17 |
| TCA                                     |                                            |      |
| Tricyclic antidepressants               |                                            |      |
| Imipramine                              | 100-200 mg/d                               |      |
| Clomipramine                            | 40-75 mg/d                                 |      |
| BZD Benzodiazepines                     |                                            |      |
| Alprazolam                              | 0.75-4.0 mg/d                              |      |
| Clonazepam                              | 0.5-2.0 mg/d                               |      |

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## CONCLUSION

Psychiatric emergencies in children and adolescents are frequent occurrences in routine clinical practice. The first contact of children and adolescents with psychiatric emergencies

may not be mental health professionals, but they are the ones contacted for an immediate consultation. Thus, all professionals dealing with children and adolescents should be equipped with the knowledge and process for the assessment and management of psychiatric emergencies in children and adolescents. Thus, this guideline provides a framework for the assessment and effective management of various psychiatric emergencies in this population and gives direction on effectively managing them. It enables the psychiatrist to ensure safety, thorough assessment and investigations, allow risk assessment, standardized care, and identify risk factors, perpetuating factors, and triggers. A step-wise approach as elucidated in the guideline allows for a comprehensive assessment and ensuring effective treatment, using both pharmacological and nonpharmacological methods. Assessment of biological, social, and psychological factors identified as a part of holistic assessment aid in formulating long-term plans and interventions for optimal outcomes. The outline of overview on assessment and management with anxiety and agitation in children and adolescents can be found in [Algorithm 5](#).

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## Conflicts of interest

There are no conflicts of interest.

## SUPPLEMENTARY FILES

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## Algorithm 1

Management of suicidal patient

[IJPsy-65-159\\_Suppl1.tif](#) (2MB, tif)

## Algorithm 2

Management of aggressive patient

[IJPsy-65-159\\_Suppl2.tif](#) (1.3MB, tif)

## Algorithm 3

Diagnostic approach to a child presenting with altered behaviour/psychotic symptoms

[IJPsy-65-159\\_Suppl3.tif](#) (3.5MB, tif)

## Algorithm 4

Management approach to a child presenting with Anxiety symptoms

[IJPsy-65-159\\_Suppl4.tif](#) (3.9MB, tif)

## Algorithm 5

Management approach to a child presenting with Anxiety symptoms

[IJPsy-65-159\\_Suppl5.tif](#) (4.8MB, tif)

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## Associated Data

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*This section collects any data citations, data availability statements, or supplementary materials included in this article.*

## Supplementary Materials

Algorithm 1

Management of suicidal patient

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## Algorithm 5

Management approach to a child presenting with Anxiety symptoms

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